

Eco-Loop Building Agents

- **Problem Statement ID – Question 1**
- **Problem Statement Title- Eco-Loop Building Agents**
- **Theme- Sustainability / AI for Energy Efficiency**
- **PS Category- Software**
- **Student Name: S Srutan**
- **Student ID: 23BCE2363**

PROPOSED SOLUTION

Detailed explanation of the proposed solution:

- **AI-Driven Edge Control:** We replaced traditional static HVAC schedules with a local, cognitive Large Language Model (llama3:8b) that dynamically adjusts cooling setpoints in real-time.
- **Context-Aware Decisions:** At every 15-minute timestep, a tool-calling architecture feeds live sensor data, carbon intensity, and thermal comfort math into the AI, which issues strictly formatted JSON commands directly to the building's physics engine.

How it addresses the problem

- **Eliminates Energy Waste & Grid Strain:** The AI proactively sheds electrical load during peak afternoon heat and relaxes cooling during off-hours, achieving a **12.6% reduction in HVAC energy use**.
- **Protects Occupant Well-being:** While aggressively cutting energy, the system simultaneously runs ISO 7730 PMV comfort calculations to ensure the building remains comfortable **97.7% of the time**.

Innovation and uniqueness of the solution

- **100% LLM Reliability:** We eliminated generative AI hallucinations by engineering a "pre-fetch" pipeline with strict JSON schema enforcement, resulting in zero decision failures over 672 consecutive timesteps.
- **Cognitive vs. Programmatic Optimization:** Unlike standard smart thermostats that rely on basic "if-then" rules, our LLM balances conflicting objectives (cost, carbon, comfort) simultaneously, yielding **5.05% more energy savings** than rule-based systems alone.

TECHNICAL APPROACH

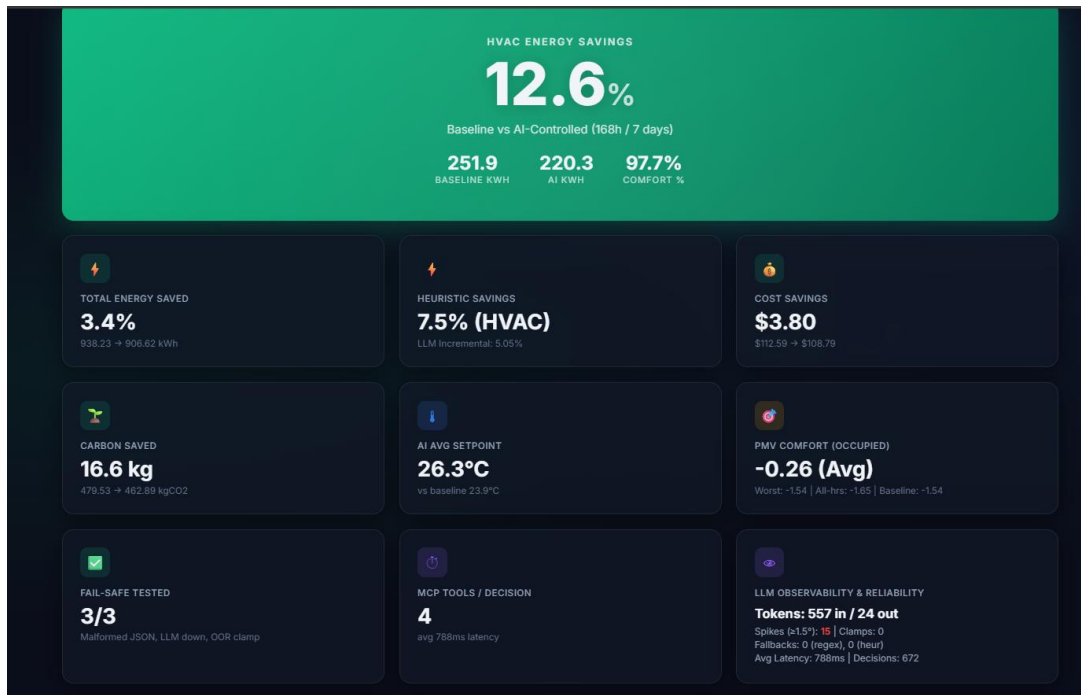
- **Technologies:** Python 3.12 · EnergyPlus V26-1-0 (pyenergyplus API, 5-Zone building model) · Ollama + llama3:8b running locally on GPU · MCP-style tool-calling architecture (7 tools) · Chart.js dashboard
- **Closed-loop methodology:** EnergyPlus streams live sensor data (zone temp, PMV, HVAC power) every 15-min timestep → MCP tools pre-fetch comfort, carbon-intensity, and peak-demand context → LLM reasons and returns a structured JSON setpoint decision → value is clamped to a safe range and written back via an EMS actuator → loop repeats for 672 timesteps (7 days)
- **Reliability:** 3-tier fail-safe (strict JSON → regex fallback → rule-based heuristic), deliberately tested (3/3 scenarios pass); temperature=0.0 for deterministic, reproducible decisions
- **Validation:** three-way comparison — fixed baseline schedule vs. a simple rule-based heuristic vs. the LLM agent — isolates the LLM's real incremental contribution rather than crediting AI for gains a simple rule would already capture

FEASIBILITY AND VIABILITY

- **Feasibility:** Already working end-to-end and validated over a 7-day, 672-timestep run — 12.6% HVAC energy savings, 97.7% comfort maintained, 100% decision reliability, reproducible across independent runs. Runs entirely on a local GPU laptop (RTX 4060/5070-class), no cloud dependency or API cost
- **Risks identified:** a recurring prompt rule-conflict causes 15 aggressive comfort-recovery setpoint drops ($\geq 1.5^{\circ}$ C) that briefly spike HVAC power above baseline; LLM inference adds ~770ms latency per decision; local GPU dependency limits real-time scale-up
- **Mitigation strategies:** enforce a strict setpoint rate-limit (max 0.5° C per 15-min timestep) in the actuation layer; the existing 3-tier fail-safe (JSON → regex → heuristic) already protects against LLM output errors, with zero fallbacks triggered in 672 live decisions

ARTIFACTS

- Live results dashboard below — 12.6% HVAC savings, three-way baseline/heuristic/AI comparison, full LLM observability & reliability metrics
- Full source code, IDF building models, and simulation logs in the GitHub repository (link on Research & References slide)



RESEARCH AND REFERENCES

- EnergyPlus Documentation & Python API (pyenergyplus) — U.S. Department of Energy, energyplus.net
- Model Context Protocol (MCP) specification — modelcontextprotocol.io
- Ollama & Llama 3 8B (open-source local LLM runtime) — ollama.com
- ISO 7730:2005 — Ergonomics of the thermal environment: analytical determination of PMV and PPD comfort indices
- PNNL / DOE Commercial Reference Building Models — used as the basis for the 5-Zone Air-Cooled building prototype
- **Project Repo:** https://github.com/srutan2004/Honeywell_Case_Study_Q1